

LKS SMK
Tingkat Nasional Ke-XXV
Tahun 2017



LKS SMK

Soal

Tingkat Nasional XXV
2017

BIDANG LOMBA

IT-Software Application



KEMENTERIAN PENDIDIKAN DAN KEBUDAYAN
DIREKTORAT JENDERAL PENDIDIKAN MENENGAH
DIREKTORAT PEMBINAAN SEKOLAH MENENGAH KEJURUAN
Kompleks Kemdiknas Gedung E Lantai 12 -13
Jalan Jenderal Sudirman Senayan Jakarta 10270
Telepon (021) 5725477 (hunting), 5725466 -69, 5725471 -75
Fax. 5725467, 5725469, 5725049
Site: www.ditpsmk.net

DESCRIPTION OF PROJECT

In this project, you are required to make an information system of Car Rental "The Blues". In General, there are three ingredients that will be given to you to develop this project, including the following:

1. ERD and Data Dictionary

This material will be used to create the database file. All entities that are created in the database must follow the ERD has been given along with the existing relationship, as well as criteria on each entity to follow a format that exists in the Data Dictionary has also been given.

2. Navigation Diagram for the Application

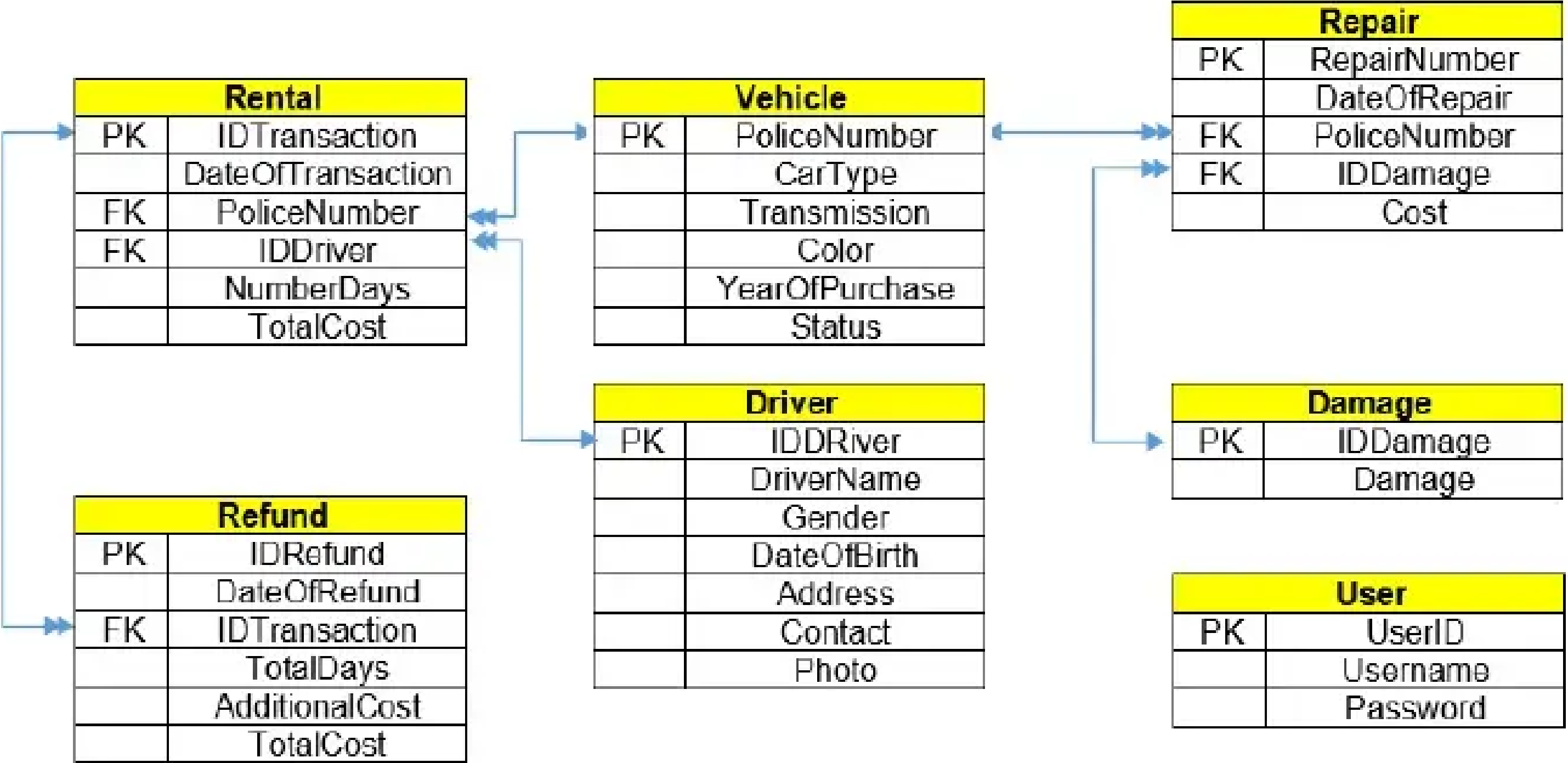
This material was made reference to create interactions between the existing form on Your project. You are free to add new interaction other than those provided without neglecting the interaction on the navigation charts that have been given.

3. Example Design (Wireframe) of Desired Information System

This material is used to guide you create the design for each form in Your project. Please note that you are not restricted in making the design of the form. Please create a design to your liking without having to be locked on the design form.

CAR RENTAL “THE BLUES” INFORMATION SYSTEM

ENTITY RELATIONSHIP DIAGRAM

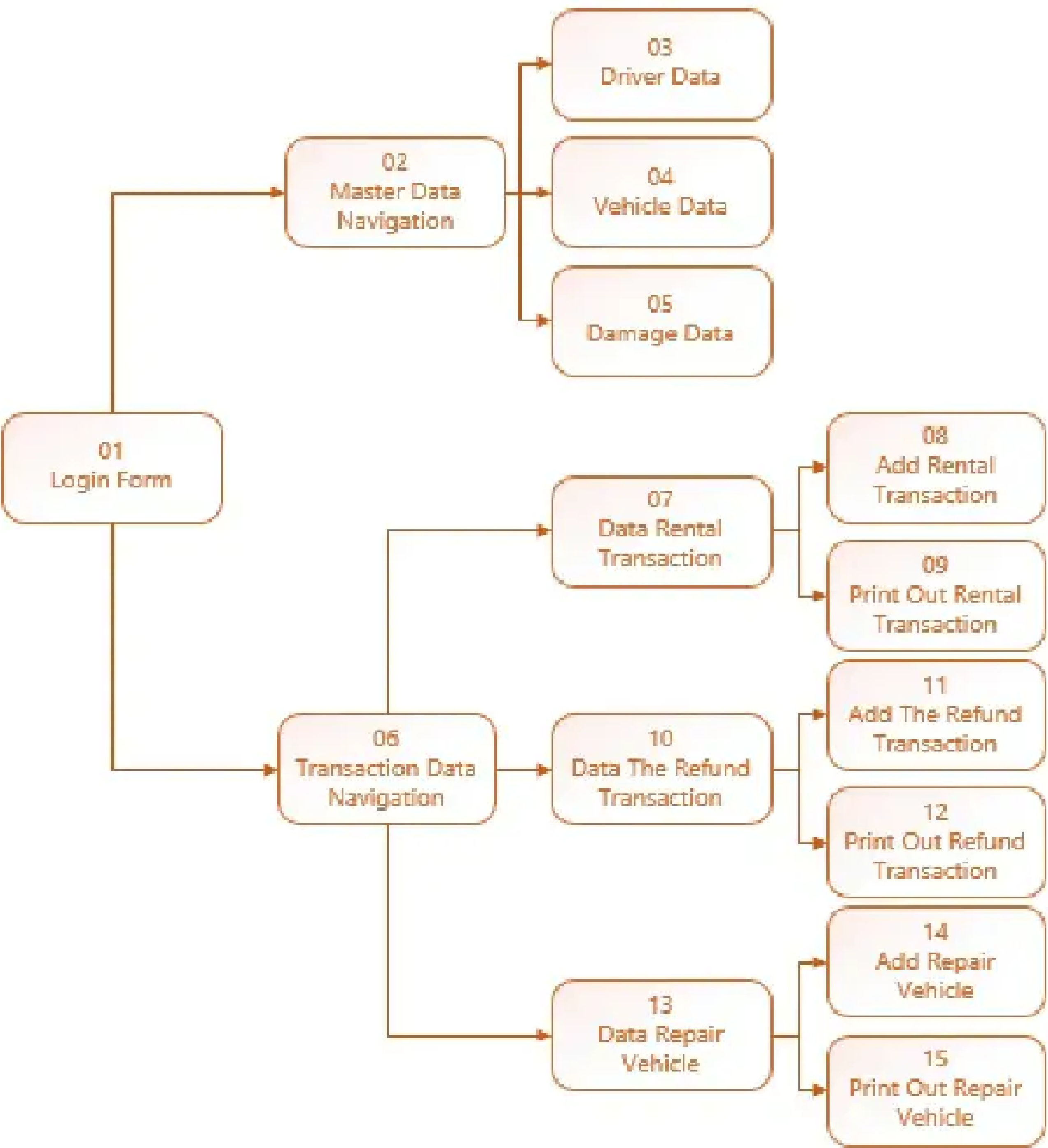


CAR RENTAL “THE BLUES” INFORMATION SYSTEM
DATA DICTIONARY

Table	Key	Field	Data Type	Required?	Notes
User	PK	UserID	INTEGER	Y	Auto Increment
		Username	VARCHAR (25)	Y	
		Password	VARCHAR(50)	Y	
Driver	PK	IDDRiver	INTEGER	Y	Auto Increment
		DriverName	VARCHAR(50)	Y	
		Gender	Boolean	Y	
		DateOfBirth	DATE	Y	
		Address	VARCHAR(100)	Y	
		Contact	VARCHAR(17)	Y	
Vehicle		Photo	VARCHAR(25)	Y	
	PK	PoliceNumber	VARCHAR(10)	Y	
		CarType	VARCHAR(50)	Y	
		Transmission	VARCHAR(9)	Y	
		Color	VARCHAR(15)	Y	
		YearOfPurchase	INTEGER	Y	
Damage		Status	VARCHAR(20)	Y	
	PK	IDDamage	INTEGER	Y	Auto Increment
Rental		Damage	VARCHAR(100)	Y	
	PK	IDTransaction	INTEGER	Y	Auto Increment
		DateOfTransaction	DATE	Y	
	FK	PoliceNumber	VARCHAR(10)	Y	
	FK	IDDriver	INTEGER	Y	
		NumberDays	INTEGER	Y	
Refund		TotalCost	DOUBLE	Y	
	PK	IDRefund	INTEGER	Y	Auto Increment
		DateOfRefund	DATE	Y	
	FK	IDTransaction	INTEGER	Y	
		TotalDays	INTEGER	Y	
		AdditionalCost	DOUBLE	Y	
Repair		TotalCost	DOUBLE	Y	
	PK	RepairNumber	INTEGER	Y	Auto Increment
		DateOfRepair	DATE		
	FK	PoliceNumber	INTEGER	Y	
	FK	IDDamage	INTEGER	Y	
		Cost	DOUBLE	Y	

CAR RENTAL “THE BLUES” INFORMATION SYSTEM

NAVIGATION DIAGRAM



CAR RENTAL “THE BLUES” INFORMATION SYSTEM
WIREFRAME

01 Login Form

Login Admin

Username :

Admin

Password :

Login

Exit

02 Master Data Navigation Form

Master Data Navigation

Welcom [User], Please select menu below

Driver Data

Vehicle Data

Vehicle Damage Data



03 Driver Data Form

Driver Data

Car Rental "The Blues"

Driver Data

ID Driver :

Name :

Gender :

Male

Female

Date of Birth :

Friday, April 14, 2017

Address :

Contact :

Browse

Save

Update

Delete

Insert New

Close

	ID Driver	Name	Gender	DateOfBirth	Address	Contact
*						

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04 Vehicle Data Form

Vehicle Data

Car Rental "The Blues"

Vehicle Data

Police Number :

Car Type :

Transmission :

Automatic

Manual

Color :

Year of Purchase :

Status :

Save

Update

Delete

Insert New

Close

	PoliceNumber	CarType	Transmission	Color	YearOfPurchase	Status
*						



05 Vehicle Damage Data Form

Vehicle Damage Data

Car Rental "The Blues"

Vehicle Damage Data

ID Damage :

Damage :

Save

Update

Delete

Insert New

Close

	IDDamage	Damage
*		

06 Transaction Data Navigation Form

Transaction Data Navigation

Welcom [User], Please select menu below

Data Rental Transaction

Data The Refund Transaction

Data Repair Vehicle



07 Data Rental Transaction Form

Data Rental Transaction

Search :

Find

Print

	IDTransaction	DateOfTransacti	PoliceNumber	CarType	IDDriver	DriverName	NumberOfDays	TotalCost
10								

Add New

Close

08 Add Data Rental Transaction Form

Rental Transaction

Car Rental "The Blues"

Rental Transaction

ID Transaction :

Date Of Transaction :

Friday , April 14, 2017

Police Number :

Car Type :

ID Driver :

Driver Name :

Number Of Day (s) :

Total Cost :

Save

Update

Delete

Insert New

Close



09 Print Out Data Rental Transaction

Car Rental “The Blues”

Data Rental Transaction

Date : April 18th 2017

ID Transaction	Date Of Transaction	Police Number	Car Type	ID Driver	Driver Name	Number of Day(s)	Total Cost
Total Cost (Rp)							

10 Data Refund Transaction

Data Refund Transaction

Search :

Find

Print

	IDRefund	DateOfRefund	IDRentalTransac	CarType	DriverName	TotalDays	AdditionalCost	TotalCost
»								

Add New

Close



11 Add Data Refund Transaction

The Refund Transaction

Car Rental "The Blues"

The Refund Transaction

ID Refund Transaction :

Date Of Refund :

Friday , April 14, 2017

ID Rental Transaction :

Police Number :

Car Type :

ID Driver :

Driver Name :

Date Of Rental :

Total Day (s) :

Additional Cost :

Total Cost :

Save

Update

Delete

Insert New

Close

12 Print Out Data Refund Transaction

Car Rental "The Blues"

Data Refund Transaction

Date : April 18th 2017

ID Refund Transaction	Date Of Refund	ID Rental Transaction	Car Type	Driver Name	Total Day(s)	Additional Cost	Total Cost
Total Cost (Rp)							



13 Data Vehicle Repair Transaction

Data Vehicle Repair

Search :

Find

Print

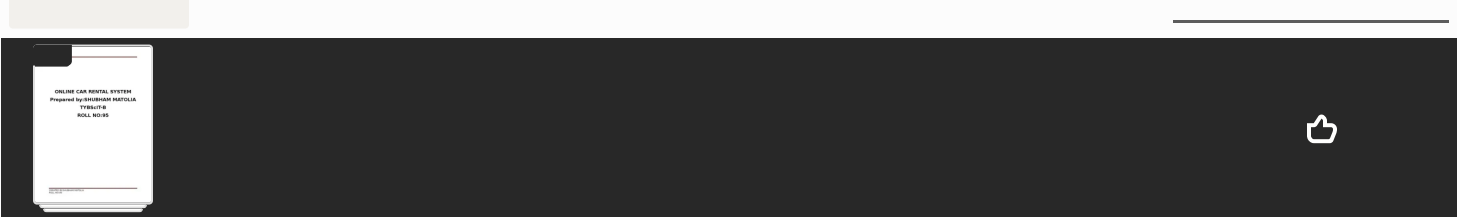
	RepairNumber	DateOfRepair	PoliceNumber	CarType	IDDamage	Damage	Cost
▶◀							

<

>

Add New

Close



14 Add Vehicle Repair Transaction

Repair Vehicle

Car Rental "The Blues"

Repair Vehicle

Repair Number :

Date Of Repair :

Friday , April 14, 2017

Police Number :

Car Type :

ID Damage :

Damage :

Cost :

Insert

Update

Delete

Clear

Close

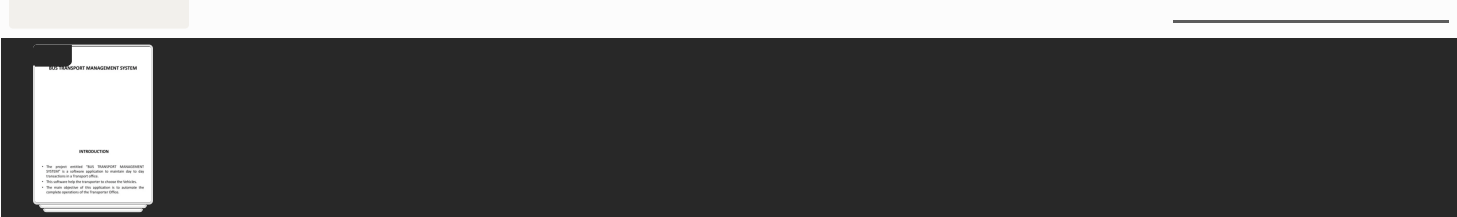
15 Print Out Data Vehicle Repair

Car Rental "The Blues"

Data Repair Print

Date : April 18th 2017

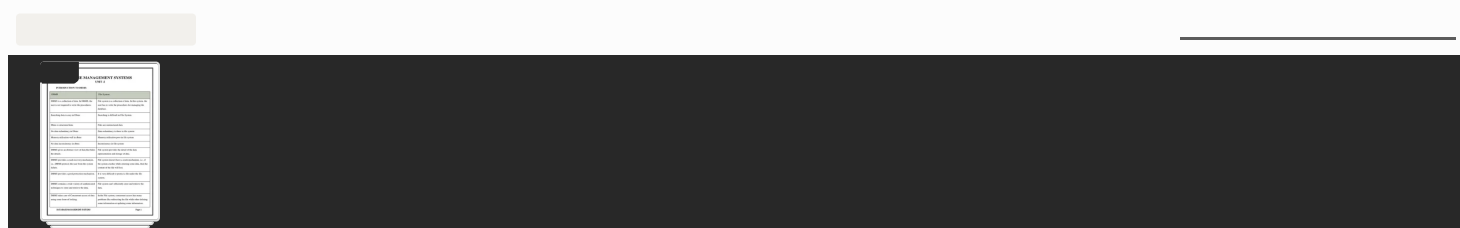
Repair Number	Date Of Repair	Police Number	Car Type	ID Damage	Damage	Cost
Total Cost (Rp)						



**THE DEVELOPMENT OF
CAR RENTAL “THE BLUES” INFORMATION SYSTEM**

A car rental company, Car Rental "The Blues" wanted to have the application that can handle all activity is in it. A process that runs on the company can be observed on Navigation Charts. Make application design, structures, as well as any requirements that need to be prepared for the company so that the transaction process business car rental company can be run well.

Using ERD, Data Dictionary, Navigation charts and the design of the Form that has been given to create an information system is needed.



Module 1 : Database Creation and Main Form Development

1. Create the Database

Create the database using your MS SQL Server on local database server following the ERD and Data Dictionary given to you.

2. Insert Data User

Add user data into the user table. Add two users in this table.

3. Create Application

Create an application using your preferred platform (C# or Visual Basic.NET).

4. Create Form “01 Login Form”

Create a login form which will be used by the Admin to be able to enter into the system. The user can used only registered users on a user table.

5. Create Form “02 Master Data Navigation”

Make a main menu for master data in accordance with the form "02 Master Data Navigation"

6. Create Form “06 Transaction Data Navigation”

Make a main menu for data transaction in accordance with the form "06 Transaction Data Navigation"

Module 2 : Master Form Development

1. Create Form “03 Driver Data”

Create a form to enter data into the master driver that will be used on the form of transactions based on form "03 Driver Data".

- All fields must be filled. A notification message will appear if there is a field that has not been filled.
- Driver-ID is automatically displayed based on the combination of sequential numbers and years. Example: 2017001 (the year 2017, the sort numbers is 001)



- Make sure the save process, edit the data, delete the data and show the data well underway
- The Data is successfully saved will go into the datagrid
- There is a search facility and photo can be displayed at the time of the search data

2. Create Form “04 Vehicle Data”

Create a form to enter data into the master of a vehicle to be used on the form of transactions based on form "04 Vehicle Data".

- All fields must be filled. A notification message will appear if there is a field that has not been filled.
- Status contains the 2 options i.e “Good Condition” and “Need Repaired”
- Make sure the save process, edit the data, delete the data and show the data well underway
- The data is successfully saved will go into the datagrid

3. Create Form “05 Damage Data”

Create a form to enter data into the master damage vehicles that will be used on the form of transactions based on form "05 Vehicle Damage Data".

- All fields must be filled. A notification message will appear if there is a field that has not been filled.
- The Damage ID is automatically displayed based on the combination of number and year sort. Example: 2017001 (the year 2017, the sort numbers 001)
- Make sure the save process, edit the data, delete the data and show the data well underway
- The data is successfully saved will go into the datagrid

Module 3 : Transaction Form and Print Out Development

1. Create Form “07 Data Rental Transaction”

Create a form to display the entire car rental based on transaction data form "07 Data Rental Transaction" with the following conditions:



- There is the facility to add new data that will show the form "08 Add Rental Transaction"
- There is a search facility based on IDTransaction and PoliceNumber
- Can print the transaction data in the form of such reports in the form "09 Print Out Rental Transaction"

2. Create Form "08 Add Rental Transaction"

Create a form to add a car rental transaction data such as form "08 Add Rental Transaction" with the following conditions:

- The IDTransaction number is incremented automatically according to the amount of data stored. IDTransaction is a combination of the number of years and sort data. Example: 2017001 (the year 2017, the sort numbers 001).
- All fields must be filled. There is a notification when there is a field that is not loaded and cannot be saved.
- The form can do new data storage, edit the data, delete the data and display the data that has been saved.
- For the Car Type and Driver Name appears automatically on the database of the reading of data from the tables Vehicle (Car Type) and a table of Driver (Driver Name)
- Only vehicle with "Good Condition" can use in transaction, not status "Need Repaired".
- Total Cost is obtained from the Number Of Day (s) multiplied by the IDR 250,000 (rent per day)

3. Create Form "09 Print Out Rental Transaction"

Create a form in the form of car rental transaction reports that can be printed to the printer as shown in the form "09 Print Out Rental Transaction"

4. Create Form "10 Data Refund Transaction"

Create a form to display the entire car returns based on transaction data form "10 Refund Transaction" with the following conditions:

- There is the facility to add a new data form that will display the "Add Refund Transaction Data 11"
- There is a data search facility based on IDRefund



- Can print the transaction data in the form of reports like form "12 Print Out Refund Transaction"

5. Create Form "11 Add Data Refund Transaction"

Create a form for adding data transaction returns a car like the form "11 Add Data Refund Transaction" with the following conditions:

- IDRefund is a number that is incremented automatically in accordance with the amount of data stored. IDRefund is a combination of the number of years and sort data. Example: 20171 (year 2017, sort number 1).
- All fields must be filled. There is a notification when there is a field that is not loaded and cannot be saved.
- The form can do new data storage, edit the data, delete the data and display the data that has been saved.
- For the Police Number, Car Type, ID Driver, Driver Name, Date Of Rental to appear automatically on the database of the reading of the data from the table Rental based on a Rental Transaction ID
- Total Day (s) appear based on the reduction between the current date to the date of rental cars.
- Additional Cost to appear in time for the purposes of borrowing money lending and are subject to an additional cost of IDR 250,000 per day
- Total Cost Total Cost is obtained from the stored on the Rental table + Additional Cost

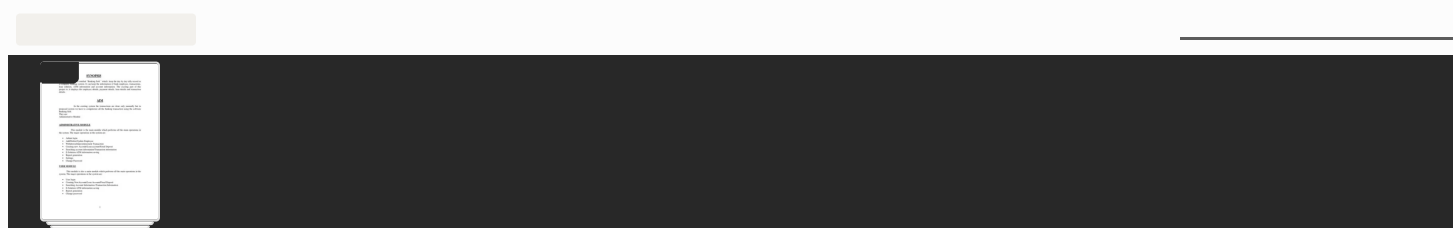
6. Create Form "12 Print Out Refund Transaction"

Create a form in the form of a car refund transaction report that can be printed to the printer as shown in the form "12 Print Out Refund Transaction"

7. Create Form "13 Data Repair Vehicle"

Create a form to display the entire repair transaction data of vehicles based on form "13 Data Repair Vehicle" with the following conditions:

- There is the facility to add new data that will show the form "14 Add Data Repair Vehicle"
- There is a data search facility based on Repair Number



- Can print the transaction data in the form of reports like form of "15 Print Out Repair Vehicle"

8. Create Form "14 Add Data Repair Vehicle"

Create a form for adding data repair vehicles such as form "14 Add Data Repair Vehicle" with the following conditions:

- Repair Number is a number that is incremented automatically in accordance with the amount of data stored. Repair Number is an combination of ordinal number and year. Example: 20171 (year 2017, sort number 1).
- All fields must be filled. There is a notification when there is a field that is not loaded and cannot be saved.
- The form can do new data storage, edit the data, delete the data and display the data that has been saved.
- For Car Type appear automatically based on the reading of data from the tables Vehicle and Damage appear automatically based on the reading of data from table Damage
- Only vehicle with "Need Repaired" can use in transaction, not "Good Condition".

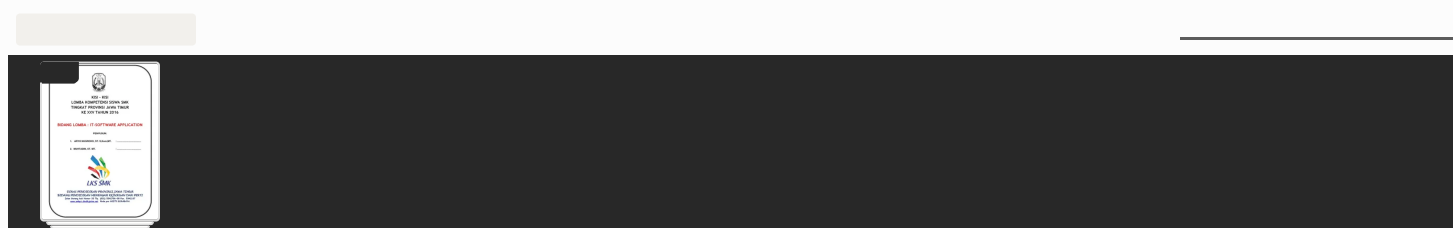
9. Create Form "15 Print Out Repair Vehicle"

Create a form in the form report vehicle repair transaction can be printed to the printer as shown in the form "15 Print Out Repair Vehicle"

Modul 4 : ERD Design and Data Dictionary

Car Rental "The Blues" would like to develop this car rental business by adding features the shopping items that can be ordered by the customer. This business allows its customers to order the desired goods, shopping without having to come directly to the store or supermarket. Shopping goods that have been ordered will be delivered directly by the driver of Car Rental "The Blues" to place the goal desired by the customer.

Shopping goods that can be ordered is the household needs, the needs of the kitchen, to the needs of others according to the wishes of the customer. There is a shopping list are booked by customers who input through the application complete with a destination address that places the shopping stuff delivered.



Another thing that has to be analyzed is the company wanted to find out the income and profits derived from transactions car rental and good service order the shopping items. In addition, the company also wanted to know whom the driver who often get the good service order, the next to be awarded a bonus when the driver gets the order at least 5 times in a month.

Based on narratives that exist, please create an ERD by using the Microsoft Visio Standard 2013 and Data Dictionary using Microsoft Office Word Professional 2013, and create table using SQL Server 2014 to support all the necessary systems in the business company Car Rental "The Blues"

Modul 5 : Result Form, ERD and Data Dictionary

1. Create “Data Table”

Create data for each table that you made in module 4. Minimal 7 data for each table

2. Create Application

Create application based on your ERD.

- Ensure your application can handle all business process in Car Rental “The Blues”
- The application must connect with your database that you have made in module 4
- Make form validation that needed in your application, for an example “all data must be filled”

